

Clustering - patient segments

DS2_Task2

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1. Introduction

Mental health disorders such as depression and anxiety are becoming increasingly common among young adults and students. Identifying patient groups with similar mental health characteristics can help healthcare professionals understand risk patterns and support targeted interventions.

This task applies K-Means clustering, an unsupervised machine learning algorithm, on mental health and demographic features to identify distinct patient profile groups.

The clustering process was performed using PHQ (Patient Health Questionnaire) scores, GAD (Generalized Anxiety Disorder) scores, and demographic-related features such as age, BMI, sleepiness score, and gender.

2. Objective

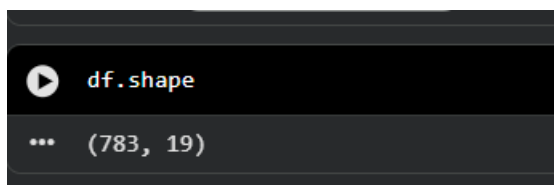
This project aimed to segment student mental health patients into meaningful profiles using unsupervised machine learning. Specifically, the goals were to:

- Apply K-Means clustering on PHQ/GAD scores and demographic features
- Identify 4–6 distinct patient clusters with clinical interpretability
- Evaluate cluster quality using silhouette score
- Generate actionable patient profiles to support mental health decision-making.

3. Dataset Description

3.1. Dataset: Depression & Anxiety Data (Kaggle) (depression_anxiety_data.csv)

The dataset contained 783 patient records across 19 columns, capturing both clinical mental health indicators and demographic characteristics of university students.



3.2. List of Columns:

```
... Index(['id', 'school_year', 'age', 'gender', 'bmi', 'who_bmi', 'phq_score',  
         'depression_severity', 'depressiveness', 'suicidal',  
         'depression_diagnosis', 'depression_treatment', 'gad_score',  
         'anxiety_severity', 'anxiousness', 'anxiety_diagnosis',  
         'anxiety_treatment', 'epworth_score', 'sleepiness'],  
        dtype='object')
```

3.3. Feature Selection:

Six features were selected for clustering based on clinical relevance and data quality:

- phq_score - PHQ depression scale (primary clinical measure)
- gad_score - GAD anxiety scale (primary clinical measure)
- age - Patient age (demographic)
- bmi - Body Mass Index (physiological)
- epworth_score - Daytime sleepiness scale (independent sleep measure)
- gender - Binary encoded (male = 1, female = 0)

Features such as depression_severity, anxiety_severity, and diagnosis flags were deliberately excluded to avoid data leakage, as they are derived from the primary PHQ/GAD scores.

```
...
```

	phq_score	gad_score	age	bmi	epworth_score	gender
0	9	11	19	33.333333	7.0	male
1	8	5	18	19.841270	14.0	male
2	8	6	19	25.102391	6.0	male
3	19	15	18	23.738662	11.0	female
4	6	14	18	25.617284	3.0	male

```
...
```

4. Data Preprocessing

Several preprocessing steps were performed before applying clustering:

- Categorical encoding: LabelEncoder applied to gender column
- Missing values: 8 rows dropped from epworth_score (1% of data)

```
... Shape after dropping nulls: (775, 6)
```

- Feature scaling: StandardScaler applied to all 6 features

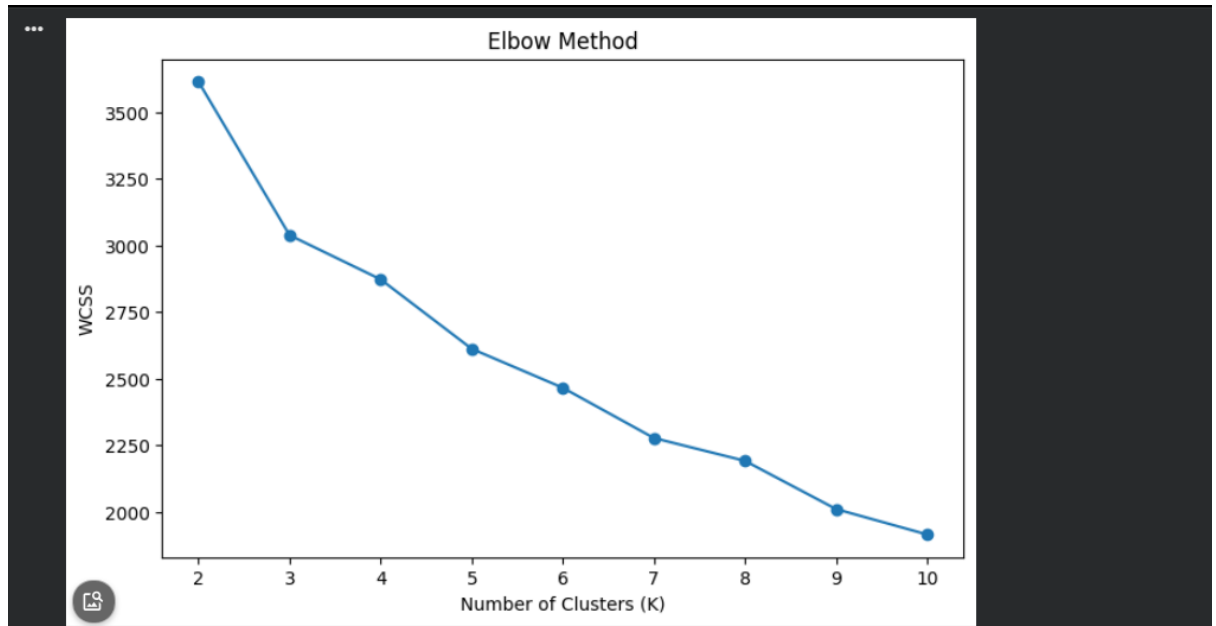
Feature scaling was important because K-Means clustering is distance-based and sensitive to feature magnitude.

5. K-Means Clustering

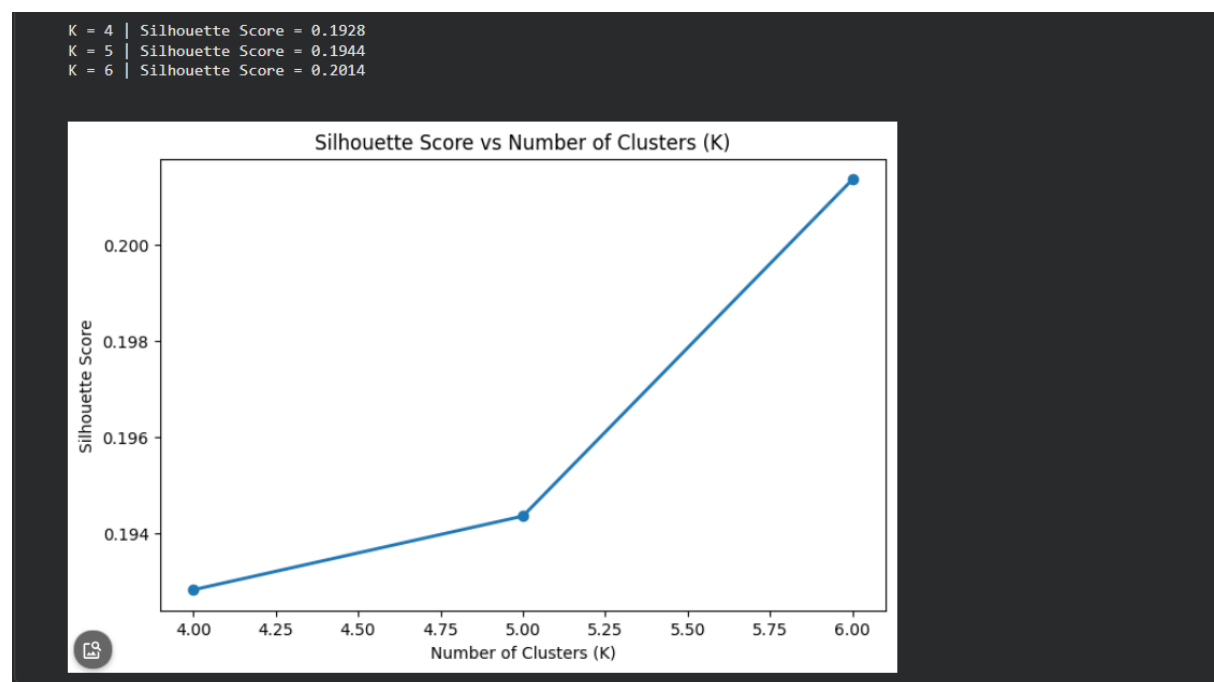
K-Means clustering was applied to group patients based on similarities in mental health indicators and demographic features.

K-Means clustering was applied with k ranging from 2 to 10. The optimal number of clusters was determined using two methods:

- Elbow Method (WCSS) - to identify the point of diminishing returns in inertia



- Silhouette Score - Silhouette scores were evaluated for k = 4, 5, and 6 as the target range specified in the task requirements.



k = 6 was selected as it produced the highest silhouette score (0.2014) within the target range. While the scores are relatively low, this is expected for mental health data where PHQ and GAD scores overlap naturally across patient groups. The elbow method further confirmed k = 6 as a reasonable inflection point.

6. Cluster Evaluation

6.1 Cluster labels

...

	phq_score	gad_score	age	bmi	epworth_score	gender	Cluster
0	9	11	19	33.333333	7.0	1	1
1	8	5	18	19.841270	14.0	1	2
2	8	6	19	25.102391	6.0	1	0
3	19	15	18	23.738662	11.0	0	4
4	6	14	18	25.617284	3.0	1	0

Next steps:

New interactive sheet

6.2. Cluster count

	count
Cluster	
0	259
5	186
2	167
3	61
1	56
4	46

dtype: int64

6.3. Cluster summary

...

	phq_score	gad_score	age	bmi	epworth_score	gender
Cluster						
0	4.81	4.30	19.92	23.86	4.86	1.00
1	8.75	7.30	19.89	31.34	5.55	0.29
2	10.13	10.98	19.73	22.22	8.61	0.20
3	6.62	5.98	23.89	25.40	6.59	0.74
4	17.17	16.15	20.07	24.48	13.13	0.46
5	4.92	4.52	20.13	20.58	4.99	0.02

Next steps:

New interactive sheet

6.4. Cluster profile

Cluster	phq_score	gad_score	age	bmi	epworth_score	gender	count	PHQ_Severity	GAD_Severity	Profile_Name	
0	0	4.81	4.30	19.92	23.86	4.86	1.00	259	Mild	Mild	Low-Risk Male Group
1	1	8.75	7.30	19.89	31.34	5.55	0.29	56	Mild	Mild	Obesity-Associated Mild Distress
2	2	10.13	10.98	19.73	22.22	8.61	0.20	167	Moderate	Moderate	Moderate Anxiety-Depression with Sleep Issues
3	3	6.62	5.98	23.89	25.40	6.59	0.74	61	Mild	Mild	Older Adult Mild Distress Group
4	4	17.17	16.15	20.07	24.48	13.13	0.46	46	Severe	Severe	High-Risk Severe Anxiety & Depression
5	5	4.92	4.52	20.13	20.58	4.99	0.02	186	Mild	Mild	Low-Risk Healthy Female Group

7. Individual Cluster Profiles

Cluster 0 - Low-Risk Male Group

This cluster represents patients mostly males with relatively low depression and anxiety scores. Sleepiness levels are also low, indicating healthy mental health conditions.

Cluster 1 - Obesity-Associated Mild Distress

Patients in this cluster show mild psychological distress along with significantly elevated BMI values. This cluster suggests a possible relationship between obesity and mental health symptoms.

Cluster 2 - Moderate Anxiety-Depression with Sleep Issues

This group shows moderate levels of anxiety and depression along with increased sleepiness scores. The findings indicate potential sleep-related mental health concerns.

Cluster 3 - Older Adult Mild Distress Group

This cluster contains comparatively older participants with mild psychological symptoms and moderate health indicators.

Cluster 4 - High-Risk Severe Anxiety & Depression

This is the most critical patient group identified in the analysis.

Key observations:

- Highest PHQ score
- Highest GAD score
- Highest sleepiness score
- Severe anxiety and depression symptoms

This cluster may require urgent psychological support and intervention.

Cluster 5 - Low-Risk Healthy Female Group

This cluster contains patients mostly females with minimal psychological distress and comparatively healthier BMI and sleep-related indicators.

8. Key findings

- 6 distinct patient profiles were successfully identified from 775 patients using K-Means clustering ($k = 6$, silhouette = 0.2014).
- Cluster 4 (High-Risk Severe) is the most critical group requiring immediate intervention only 46 patients but with PHQ 17.17 and GAD 16.15.
- Cluster 2 (Moderate Mixed) is the largest at-risk group with 167 patients showing moderate symptoms and sleep issues.
- BMI emerged as a key differentiator - Cluster 1 had mean BMI of 31.34 despite only mild mental health symptoms.
- Gender was a strong separating feature - Clusters 0 and 5 are near-exclusively male and female respectively.
- Sleep quality (Epworth score) correlates strongly with symptom severity - Cluster 4 had the highest Epworth score (13.13).
- Age distinguishes Cluster 3 - older students (mean 23.89) show a distinct mild distress pattern compared to younger ones.

9. Limitations:

While this analysis successfully identified six meaningful patient clusters, several limitations should be acknowledged:

9.1. Low Silhouette Score (0.2014)

The silhouette score of 0.2014 indicates moderate cluster separation. This is expected in mental health data where PHQ and GAD scores naturally overlap across patient groups - clear-cut boundaries between mild, moderate, and severe symptoms rarely exist in clinical reality. The clusters should be interpreted as tendencies rather than rigid classifications.

9.2. Small High-Risk Cluster

Cluster 4 (High-Risk Severe Anxiety & Depression) contains only 46 patients approximately 6% of the dataset. While clinically the most critical group, its small size limits the statistical confidence of its profile and may not be fully representative of the broader high-risk population.

9.4. K-Means Algorithm Constraints

K-Means assumes spherical, equally sized clusters and is sensitive to outliers. Mental health data is rarely distributed in this way. Alternative algorithms such as DBSCAN or Gaussian Mixture Models may capture more complex cluster shapes and could be explored in future work.

9.5. Cross-Sectional Data

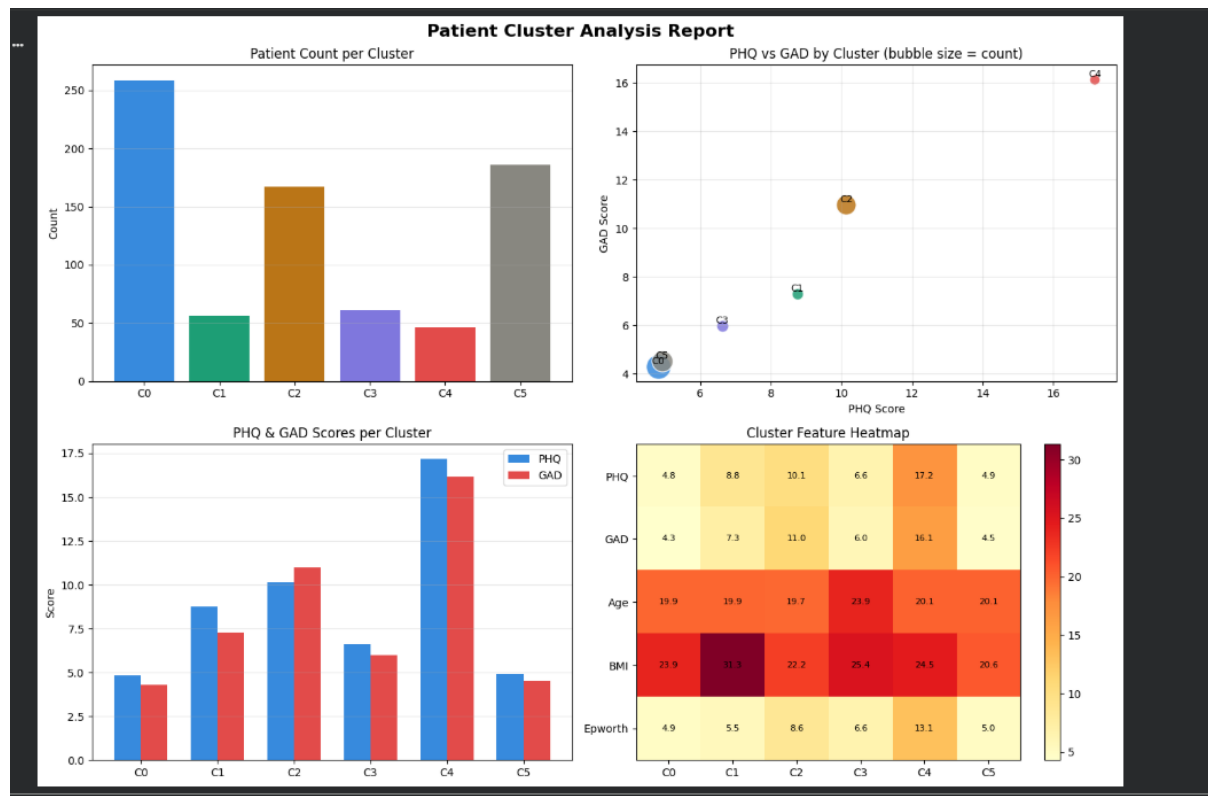
The dataset represents a single point-in-time snapshot. Mental health conditions are dynamic and can change significantly over weeks or months. Longitudinal data would provide more reliable and actionable patient profiles.

10. Visualization Analysis

A cluster dashboard was created to visually analyze patient segmentation.

The dashboard includes:

- Patient count per cluster
- PHQ vs GAD scatter analysis
- PHQ and GAD comparison charts
- Cluster feature heatmap



11. Conclusion

This project demonstrated the effectiveness of K-Means clustering for mental health patient segmentation.

Using PHQ, GAD, and demographic features, six meaningful patient profiles were identified. The clustering results highlighted different mental health severity levels, demographic patterns, sleep-related issues, and obesity-associated distress patterns.

The findings from this analysis may help support targeted mental health interventions, early risk identification, and patient profiling strategies in healthcare analytics.

12. Deliverables

- [clustered_patients.csv](#)
- [cluster_profiles.csv](#)
- [Silhouette Score Graph](#)
- [Elbow Method Graph](#)
- [Cluster Dashboard Visualization](#)